

A Propensity Score-Inspired Randomness Test for Binary Sequence Analysis

ABSTRACT

We test for randomness in binary sequences using the **Rubin Causal Model** and propensity scores. By matching blocks with similar scores and comparing their outcomes, our method reveals hidden dependencies that standard randomness tests may miss. True random generators pass this test, while generators with subtle structure are flagged by significant deviations in our statistics.

INTRODUCTION



The **Rubin Causal Model**, which was invented by Donald Rubin in the 1970s, makes causal effect estimates in terms of potential outcomes by comparing what did happen to what would have happened had subjects received different treatments. The model determines relationships between patient characteristics and their causal effects through the representation of treatment responses and patient characteristics in bit-strings. If two patients are similar on treatment or no treatment, their similar characteristics can serve as a platform for predicting causal effects in another patient. The platform uses a propensity score, calculated from patient characteristics, to match individuals with similar probabilities of receiving treatment. Differences in their characteristics offer the experimental platform for estimating causal effects.

Example of Potential Outcomes, Causal Effect, Actual Treatment, and Observed Outcome¹

Potential Outcomes (Not Observable)			Known		
Unit	<i>Y(Aspirin)</i>	<i>Y(No Aspirin)</i>	Causal Effect	Actual Treatment	Observed Outcome
You	No Headache	Headache	Improvement due to Aspirin	Aspirin	No Headache

¹ Imbens, G.W. & Rubin, D.B. (2015). Causal Inference for Statistics, Social, and Biomedical Sciences: An Introduction. Cambridge University Press.

HYPOTHESES

Null Hypothesis (H₀)

The input bit sequence is random. After matching blocks by propensity scores (logistic for single treatment bit, multinomial for multiple), matched pairs will be independent of treatment assignment, and the average Hamming distance over outcome bits will be 0.5 (random chance).

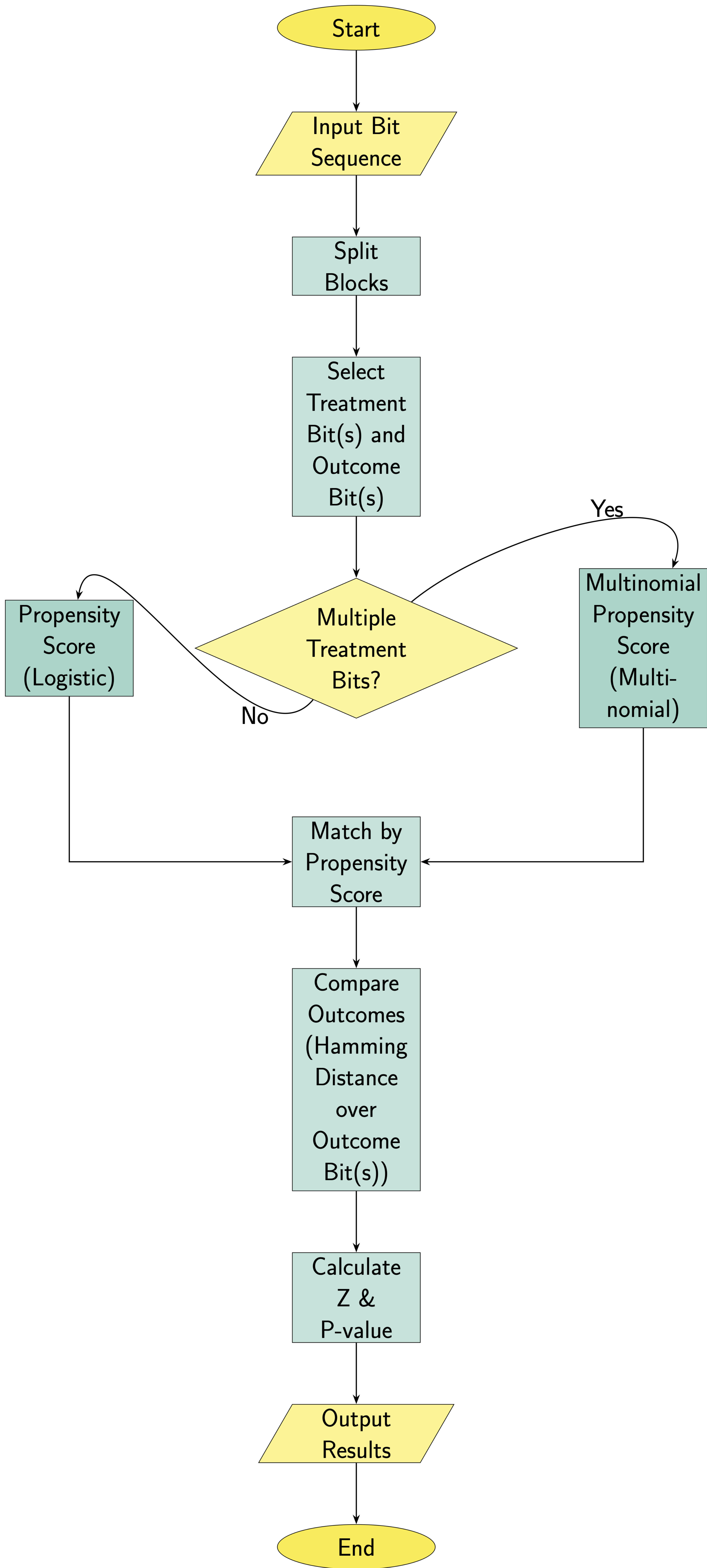
Alternative Hypothesis (H₁)

The input bit sequence is not random. There is dependency or structure between treatment and outcome bits. After matching on propensity scores, the observed average Hamming distance between matched pairs' outcome bits

HYPOTHESIS TESTING FRAMEWORK

For the **Rubin Causal Model** test, we input bit sequence is split into blocks, treatment and outcome bits selected, propensity score model (logistic or multinomial) is fit, matching and comparison performed, then Z-score and p-value are reported.

HYPOTHESIS TESTING ALGORITHM



HYPOTHESIS TESTING EVALUATION

Mersenne Twister: Classic pseudo-random generator, widely used in simulations.
NumPy PCG64: Modern, statistically strong pseudo-random generator (NumPy default).
os.urandom: Operating system–based cryptographically secure RNG.
secrets: High-security RNG for cryptography (Python’s secrets module).
LFSR: Linear Feedback Shift Register. Simple hardware-style generator, not cryptographically secure.
Patterned: Artificial sequence with a fixed, non-random pattern (control).
NIST: Official bitstream from the National Institute of Standards and Technology Statistical Test Suite

RESULTS

Rubin Randomness Test: NIST Random Generated Number Sequence (Single Treatment/Outcome)

Sequence	Subsequence Size	Treatment Positions	Outcome Positions	Total Matches	Average Distance	Z-score	P-value	Interpretation
NIST Random Generated Number Sequence (500,000 bits)	10	[2]	[7]	24992	0.501	0.43	0.667	Fail to reject H ₀ (random)

Rubin Randomness Test: NIST Random Generated Number Sequence (Multi-Treatment/Outcome)

Sequence	Subsequence Size	Treatment Positions	Outcome Positions	Total Matches	Average Distance	Z-score	P-value	Interpretation
NIST Random Generated Number Sequence (500,000 bits)	16	[2, 5, 9]	[7, 10, 12]	3978	1.497	-0.127	0.899	Fail to reject H ₀ (random)

Rubin Randomness Test: Generator Comparison (Single Treatment/Outcome)

Generator	Subsequence Size	Treatment Positions	Outcome Positions	Total Bits	Average Distance	Z-score	P-value	Interpretation
Mersenne Twister	10	[2]	[7]	500,000	0.501	0.247	0.805	Fail to reject H ₀ (random)
NumPy PCG64	10	[2]	[7]	500,000	0.502	0.683	0.494	Fail to reject H ₀ (random)
os.urandom	10	[2]	[7]	500,000	0.500	-0.025	0.980	Fail to reject H ₀ (random)
secrets	10	[2]	[7]	500,000	0.500	0.146	0.884	Fail to reject H ₀ (random)
LFSR (random seed)	10	[2]	[7]	500,000	0.498	-0.589	0.556	Fail to reject H ₀ (random)
Patterned (control)	10	[2]	[7]	500,000	1.000	158.114	0.000	Reject H ₀ (not random)

Rubin Randomness Test: Generator Comparison (Multi-Treatment/Outcome)

Generator	Subsequence Size	Treatment Positions	Outcome Positions	Total Bits	Average Distance	Z-score	P-value	Interpretation
Mersenne Twister	16	[2, 5, 9]	[7, 10, 12]	500,000	1.496	-0.169	0.866	Fail to reject H ₀ (random)
NumPy PCG64	16	[2, 5, 9]	[7, 10, 12]	500,000	1.481	-0.781	0.435	Fail to reject H ₀ (random)
os.urandom	16	[2, 5, 9]	[7, 10, 12]	500,000	1.492	-0.333	0.739	Fail to reject H ₀ (random)
secrets	16	[2, 5, 9]	[7, 10, 12]	500,000	1.501	0.032	0.975	Fail to reject H ₀ (random)
LFSR (random seed)	16	[2, 5, 9]	[7, 10, 12]	500,000	1.548	2.001	0.045	Reject H ₀ (not random)
Patterned (control)	16	[2, 5, 9]	[7, 10, 12]	500,000	3.000	125.000	0.000	Reject H ₀ (not random)

CONCLUSION AND FUTURE WORK

Our model is successful because it detects dependencies that classical tests overlook. Next, we plan to extend our method by integrating Pearl’s causal framework for more comprehensive analysis.